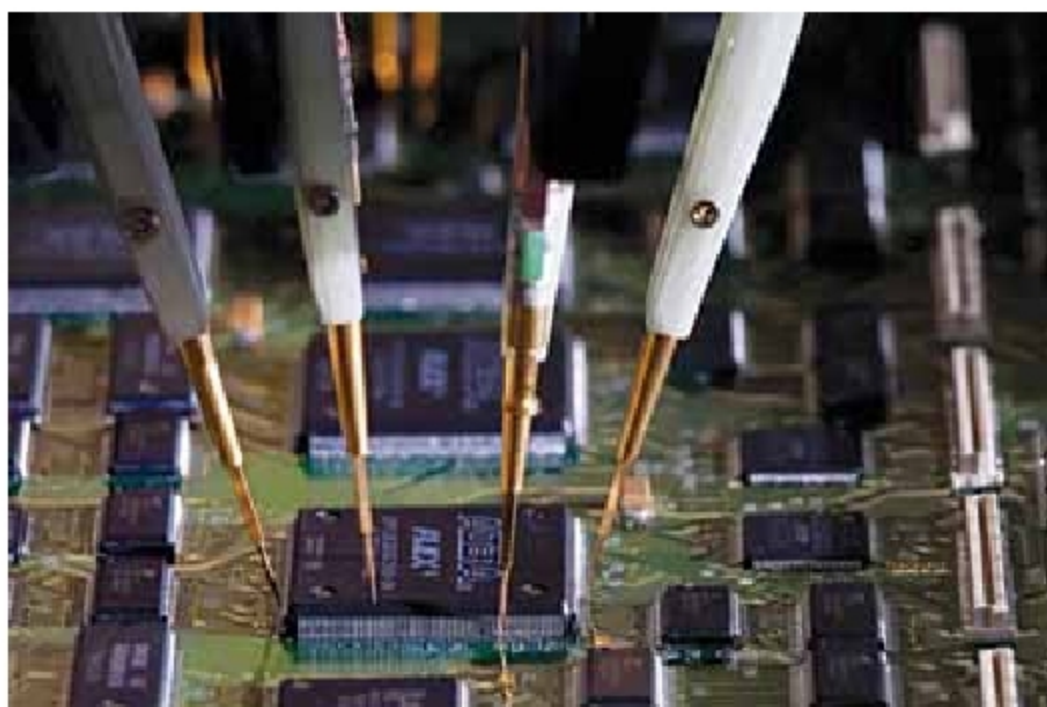




How To Build An Automated Test Jig To Check PCBs

Printed circuit board testing is an important task for test engineers. However, with increase in complexity of the circuit, it has become difficult to test it. So, nowadays, most of the industries are using a fully automatic test jig that can run the test and make an automated test report



(Credit: <https://i1.wp.com/www.onethesis.com>)



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Printed circuit board (PCB) testing is an inevitable task for test engineers. With PCB size becoming smaller and complexity of the circuit increasing, it has become difficult for engineers to perform testing. With a large number of PCBs to test, keeping records manually is not possible. Moreover, sharing all documents is also a difficult task as, generally, testing is not done by original equipment manufacturers (OEMs).

One solution that most of the industries are now using is a simple bed of nails test jig in which a test engineer has to select the switches manually and note down the test point values. But doing things manually can cause errors.

So, the solution to the above problem is to use a fully automatic test jig that can run the test and make an automated test report.

To build an automated test jig use a bed of nails test jig and connect all test points of the device under test (DUT) to the data acquisition board (DAQ). This process will acquire all test points as per the test specified, apply some pass-fail criteria, and finally give a report. DAQ board can be fitted inside the test jig.

The DAQ board runs using an automat-

ed software like LabVIEW or Visual Basic. It can be customised as per requirements, or a standard DAQ from the market can be used.

Data for test jig manufacture

The data required for manufacturing test jigs are as follows:

- Gerber files of the device under test
- List of test points and their positions. Normally, through-hole test points are used, but in some cases, SMD points can also be taken
- DAQ board size with connectivity drawings
- Any other PCBs required for testing DUT
- Cables coming in or out of the jig
- Power supply for DUT
- Connectors for the test jig

Automated software

There are other software as well for automating the test, but the most commonly used software programs are LabVIEW and Visual Basic.

LabVIEW. It is a software made by NI, formerly known as National Instruments Corporation. It uses a graphical programming language. With some knowledge, it